Daily Combinatorics Drill #2

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Combinations $\binom{n}{r}$ · Choosing vs Arranging · Complementary Counting · Committee Constructions.

Section A Rapid Recognition ($\leq$15 s each)

1. Evaluate: $\binom{7}{2}$.

Ans: 21

2. Evaluate: $\binom{8}{3}$.

Ans: 56

3. Evaluate: $\binom{9}{7}$.

Ans: 36

4. How many committees of 3 can be chosen from 6 people?

Ans: $\binom{6}{3} = 20$

5. Solve for n: $\binom{n}{2} = 45$.

Ans: $n = 10$

6. How many subsets of a 5-element set have at least 4 elements?

Ans: $\binom{5}{4} + \binom{5}{5} = 5 + 1 = 6$

7. Evaluate: $\binom{6}{0} + \binom{6}{1} + \cdots + \binom{6}{6}$.

Ans: $2^6 = 64$

8. How many ways can 2 letters be chosen from {A, B, C, D, E}?

Ans: $\binom{5}{2} = 10$

9. True or false: $\binom{100}{98} = 4950$.

Ans: True

10. A lottery ticket selects 4 numbers from 1–10 (order irrelevant). How many different tickets are there?

Ans: $\binom{10}{4} = 210$

Section B Manipulation Drills ($\leq$50 s each)

1. Solve for n: $\binom{n}{3} = 35$.

Ans: $n = 7$

2. From 8 men and 6 women, how many committees of 2 men and 2 women can be formed?

Ans: $\binom{8}{2}\binom{6}{2} = 28 \cdot 15 = 420$

3. Twelve people at a party all shake hands with each other exactly once. How many handshakes take place?

Ans: $\binom{12}{2} = 66$

4. Ten points, no three collinear: how many segments, and how many triangles?

Ans: $\binom{10}{2} = 45$ segments; $\binom{10}{3} = 120$ triangles

5. Team of 5 from 11, then a captain from the team. Count both ways.

Ans: $\binom{11}{5} \cdot 5 = 2310 = 11 \cdot \binom{10}{4}$

6. How many binary strings of length 8 contain exactly three 1s?

Ans: $\binom{8}{3} = 56$

7. From the 5 vowels and 21 consonants, how many 3-letter *sets* have exactly one vowel and two consonants?

Ans: $\binom{5}{1}\binom{21}{2} = 5 \cdot 210 = 1\,050$

8. Solve for n: $\binom{n}{2} + \binom{n}{1} = 21$.

Ans: $n = 6$

9. Committees of 4 from 9 people: how many include Priya? Exclude her?

Ans: Include: $\binom{8}{3} = 56$; exclude: $\binom{8}{4} = 70$

10. Routes from $(0,0)$ to $(4,3)$ stepping only right or up?

Ans: $\binom{7}{3} = 35$

Section C Substitution & Structure ($\leq$90 s each)

1. Committee of 5 from 7 women and 6 men, at least 4 women.

Ans: $\binom{7}{4}\binom{6}{1} + \binom{7}{5} = 35 \cdot 6 + 21 = 231$

2. How many four-digit numbers have at least one repeated digit?

Ans: $9000 - 9 \cdot 9 \cdot 8 \cdot 7 = 9000 - 4536 = 4464$

3. How many diagonals does a regular 12-gon have?

Ans: $\binom{12}{2} - 12 = 66 - 12 = 54$

4. Triangles from 8 points on a circle; how many avoid point A?

Ans: $\binom{8}{3} = 56$ in total; $\binom{7}{3} = 35$ avoid A

5. Teams of 6 from 6 boys and 6 girls with unequal numbers of each.

Ans: $\binom{12}{6} - \binom{6}{3}^2 = 924 - 400 = 524$

6. 5-card hands from 52 cards containing at least three aces.

Ans: $\binom{4}{3}\binom{48}{2} + \binom{48}{1} = 4512 + 48 = 4\,560$

7. A convex n-gon has exactly 27 diagonals. Find n.

Ans: $n = 9$

8. Subsets of $\{1, \ldots, 10\}$ containing at least one of 1 and 2.

Ans: $2^{10} - 2^8 = 1024 - 256 = 768$

Section D Challenge (SMC difficulty)

1. Rectangles on a 4×4 grid. *(extends Sheet 1, D2)*

Ans: $\binom{5}{2}^2 = 100$

2. Subsets of $\{1, \ldots, 8\}$ with no two consecutive integers (empty set included). *(cf. MAT 2023 Q5)*

Ans: 55

3. Committees of any size from 6 juniors and 6 seniors with strictly more juniors.

Ans: $\dfrac{2^{12} - \binom{12}{6}}{2} = \dfrac{4096 - 924}{2} = 1\,586$

4. Six people each shake hands with exactly two others. In how many ways?

Ans: 70

5. A committee of 4 is chosen from 9 people (5 women including Priya, 4 men). Consider statements: (A) Priya is on the committee, (B) exactly 2 women are on the committee, (C) at least 3 men are on the committee. Exactly one of the three statements is true. How many committees satisfy this? *(after TMUA Practice Paper 2 Q4)*

Ans: 81